Assignment No. 13

15.08.20216

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Objective

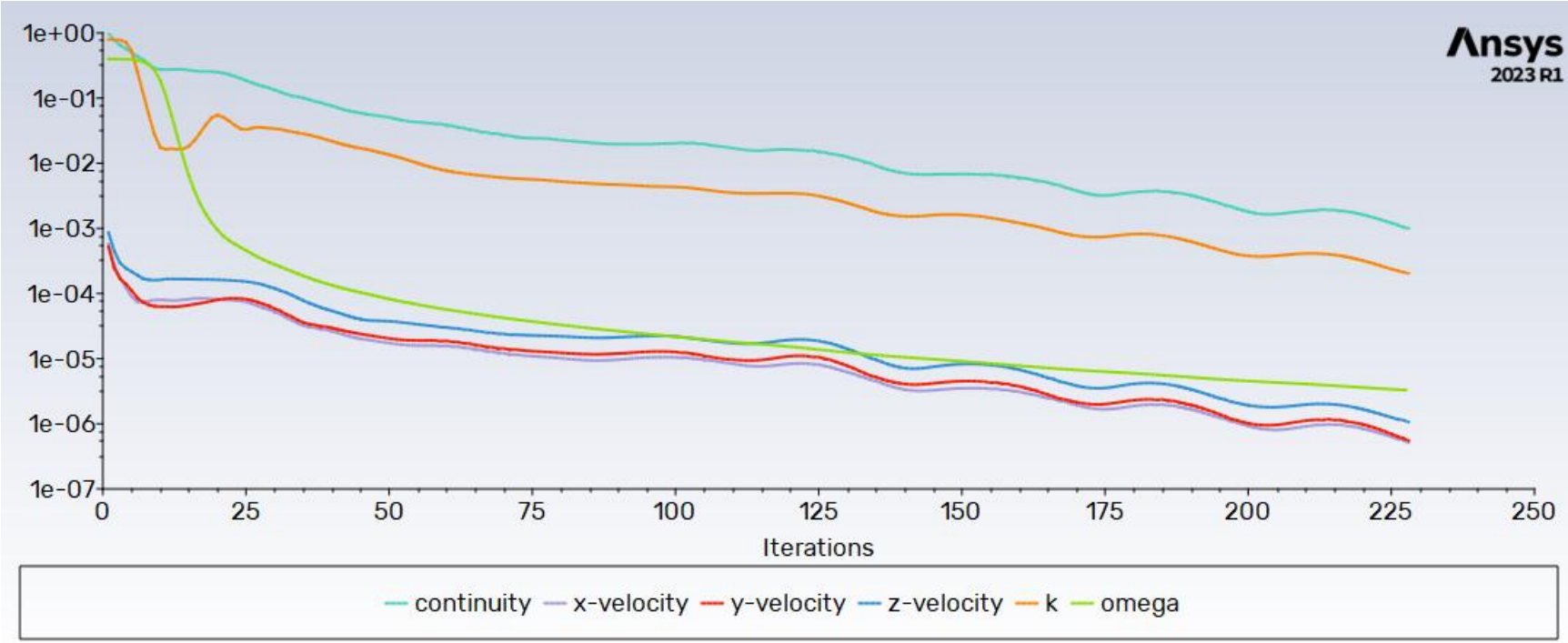
Comparative study: SS and transient flow over a cylinder of 1m diameter and 1 meter length

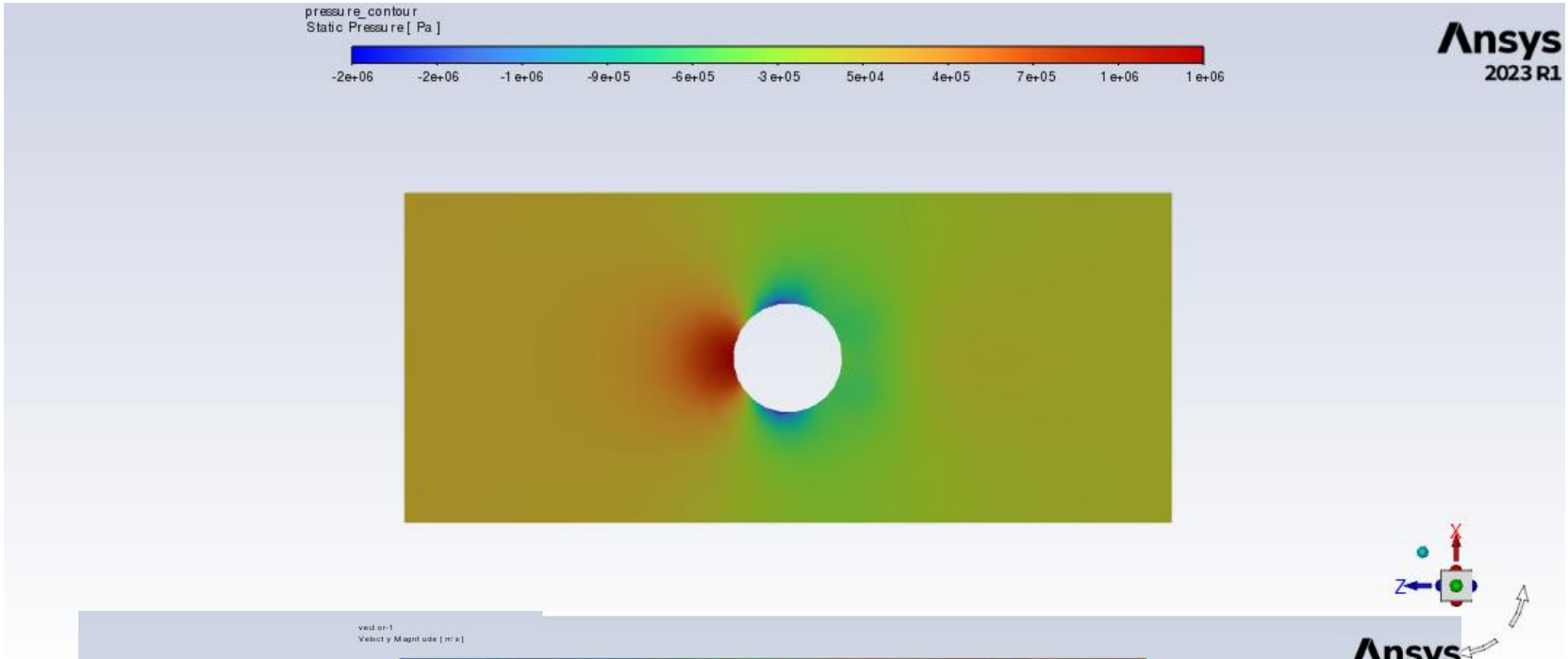
Problem Statement

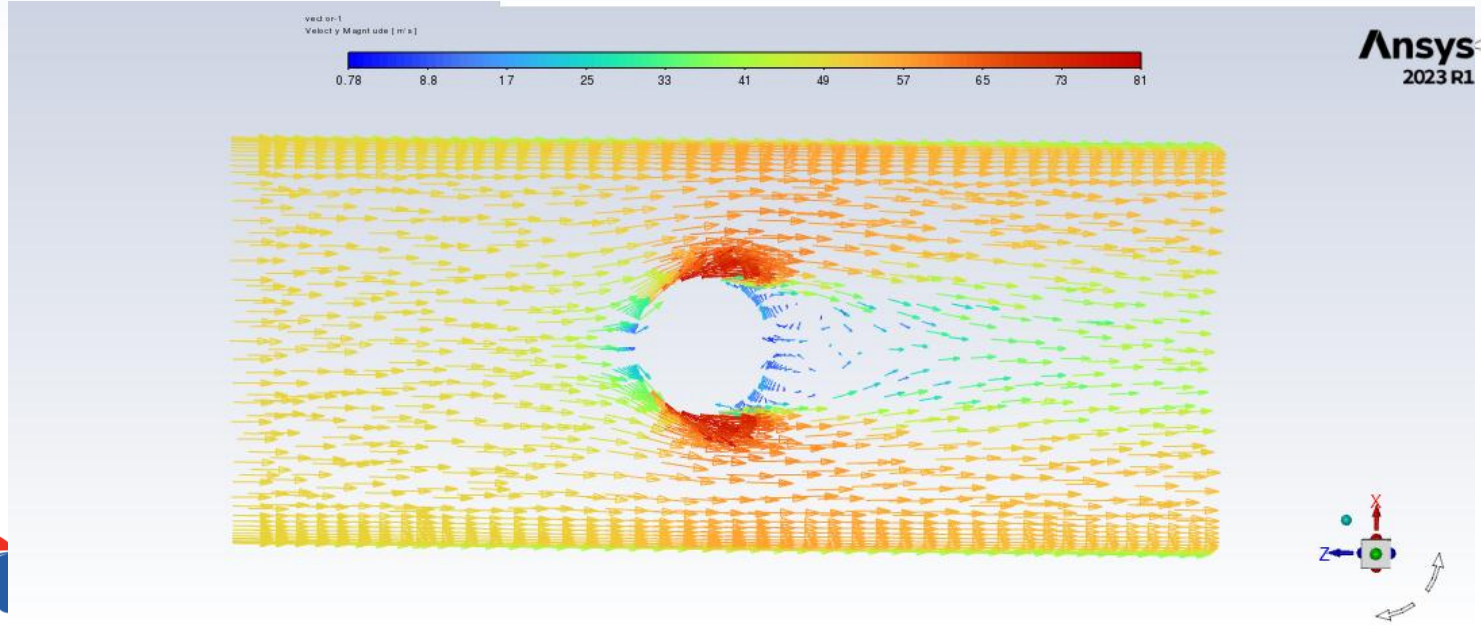
- Inlet velocity: 50 m/s
- Compare Cd for both cases
- Set up a pressure and velocity monitor at the wake and perform FFT on the temporal variation of pressure and velocity

Steady State

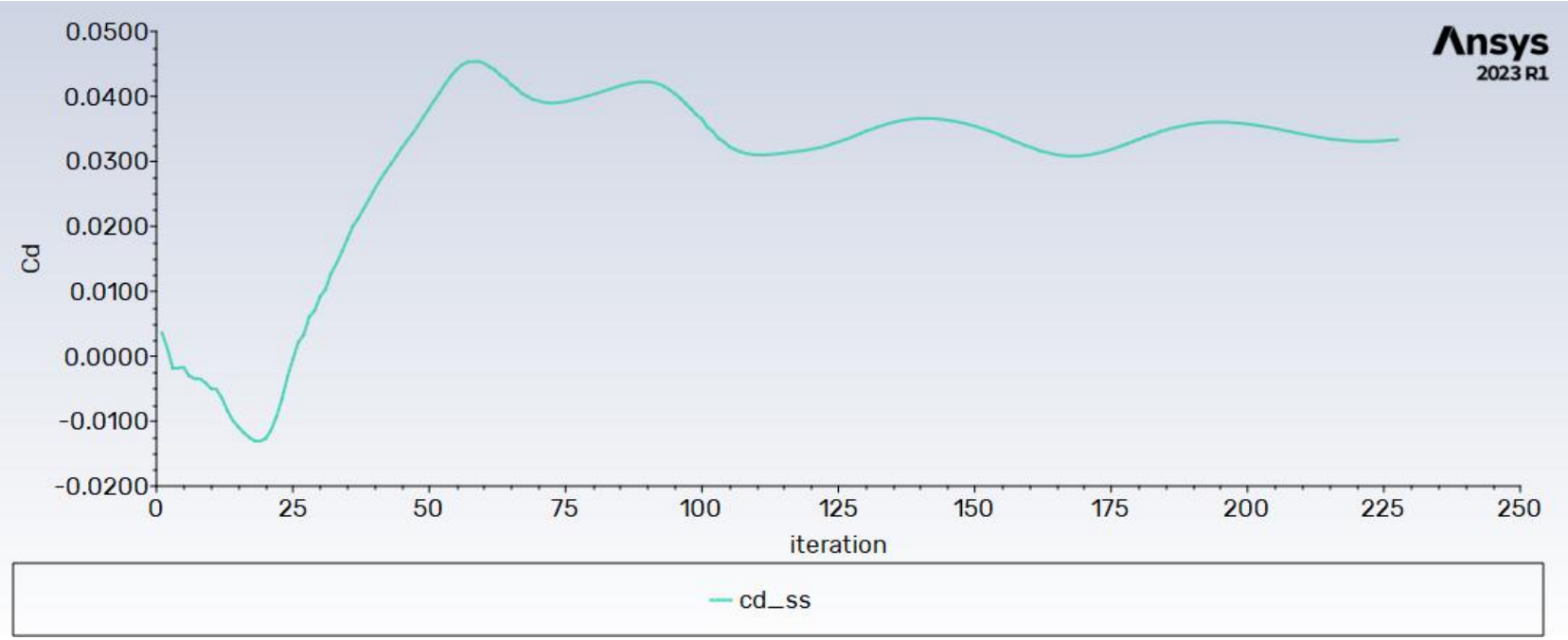
Used Under relaxation factor




pressure_contour
Static Pressure [Pa]
-2e+06
-2e+06
-1e+06
-9e+05
-6e+05
-3e+05
5e+04
4e+05
7e+05
1e+06
1e+06
Ansys
2023 R1
X
Z


vector-1
Velocity Magnitude [m/s]
0.78
8.8
17
25
33
41
49
57
65
73
81
Ansys
2023 R1
X
Z

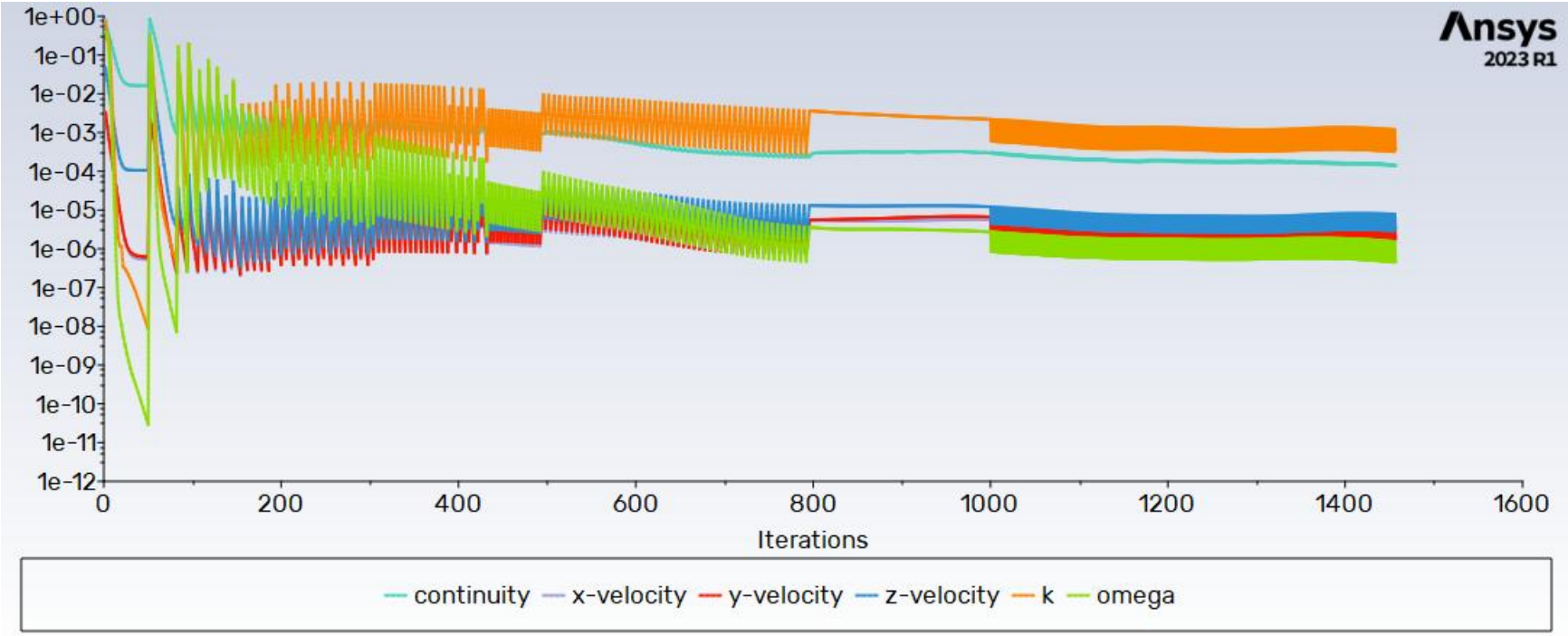
Cd is roughly 0.031

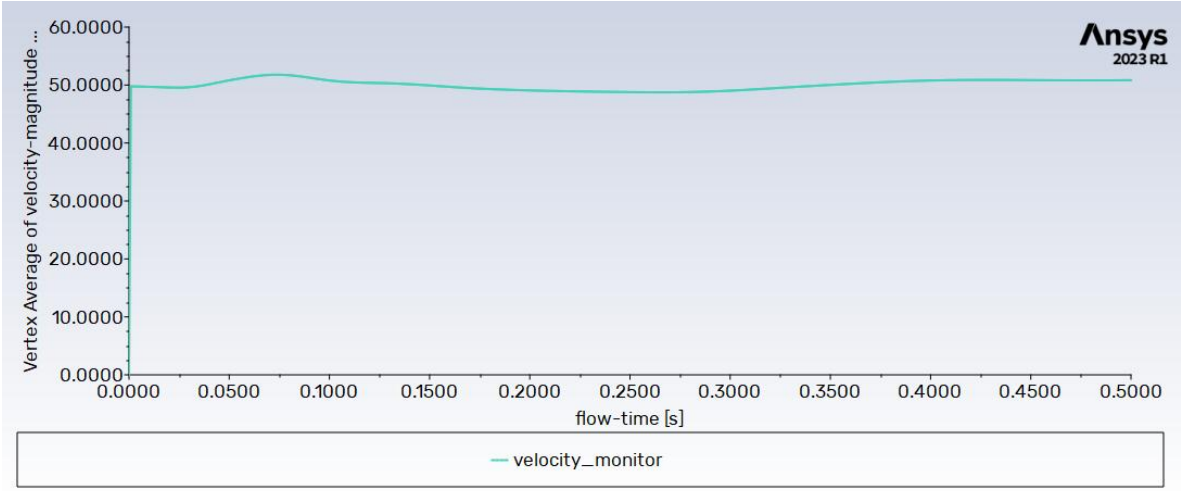


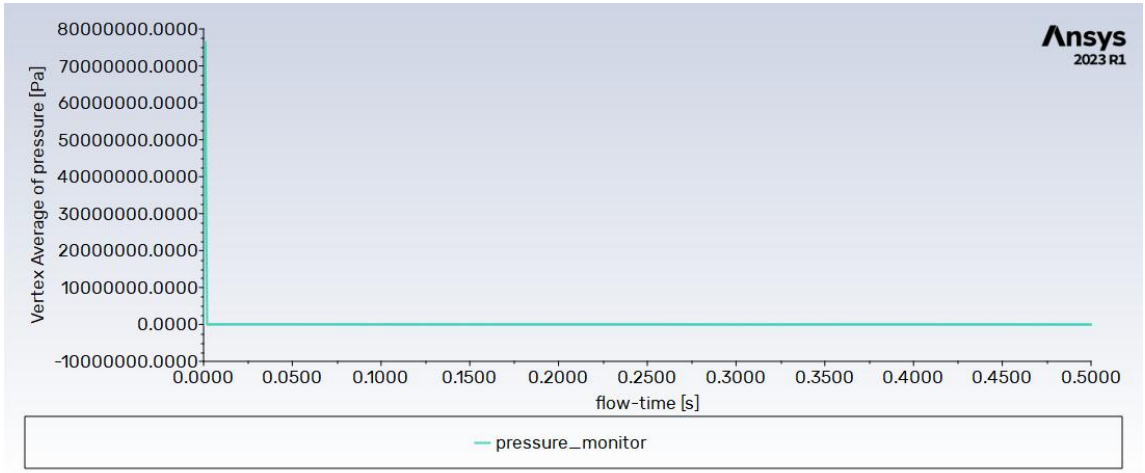
Transient

Time step size used: 7(length scale)/50~0.9

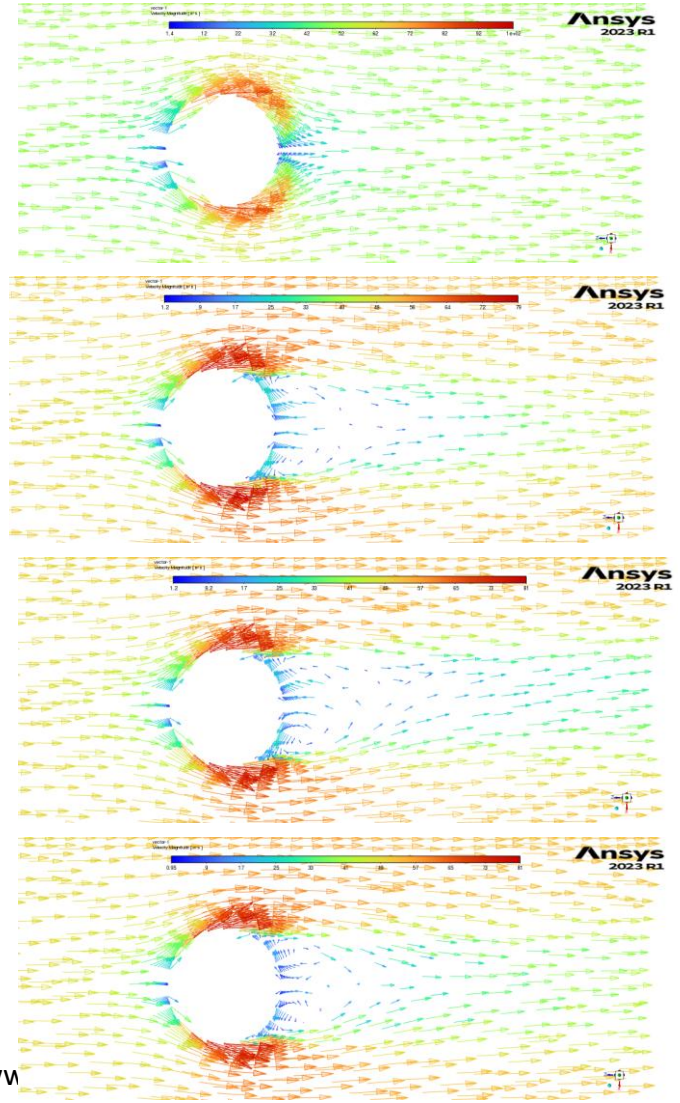
Total flow time: 0.5 sec



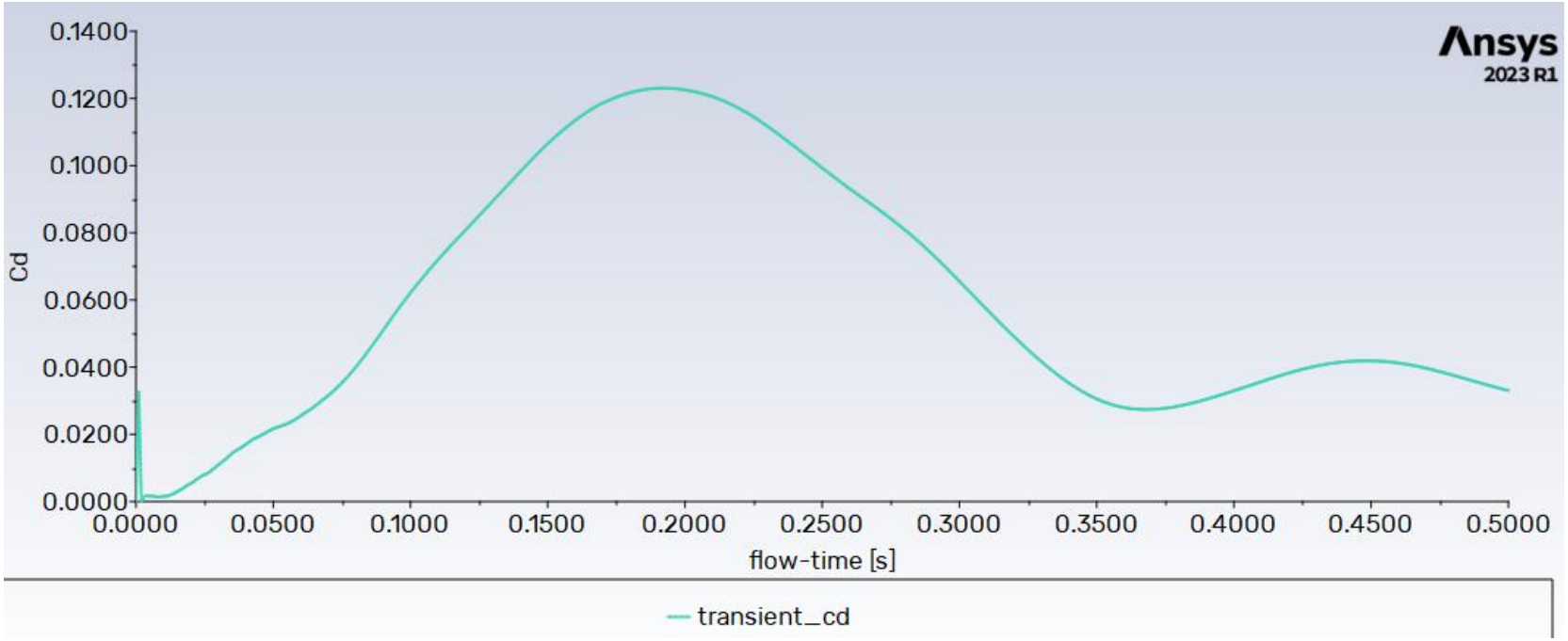

Ansys
2023 R1
Vertex Average of velocity-magnitude ...
60.0000
50.0000
40.0000
30.0000
20.0000
10.0000
0.0000
0.0000
0.0500
0.1000
0.1500
0.2000
0.2500
0.3000
0.3500
0.4000
0.4500
0.5000
flow-time [s]
velocity_monitor


Ansys
2023 R1
Vertex Average of pressure [Pa]
80000000.0000
70000000.0000
60000000.0000
50000000.0000
40000000.0000
30000000.0000
20000000.0000
10000000.0000
0.0000
-10000000.0000
0.0000
0.0500
0.1000
0.1500
0.2000
0.2500
0.3000
0.3500
0.4000
0.4500
0.5000
flow-time [s]
pressure_monitor

Some velocity contours(per 100 timesteps)



Cd value was roughly 0.04



Conclusion

- Cd value was close for both cases. FFT wasn't performed as the pressure and velocity didn't seem to vary at that specific point or the transient solution was incorrect,
- But it can be concluded that Cd sits around 0.035

Appendix

- Include the program (MATLAB code or UDFs etc) here